

SESSION LAYER

The Session layer is the fifth layer in the OSI (Open Systems Interconnection) model. It plays a crucial role in managing and controlling the dialog between two computers, establishing, maintaining, and terminating communication sessions.

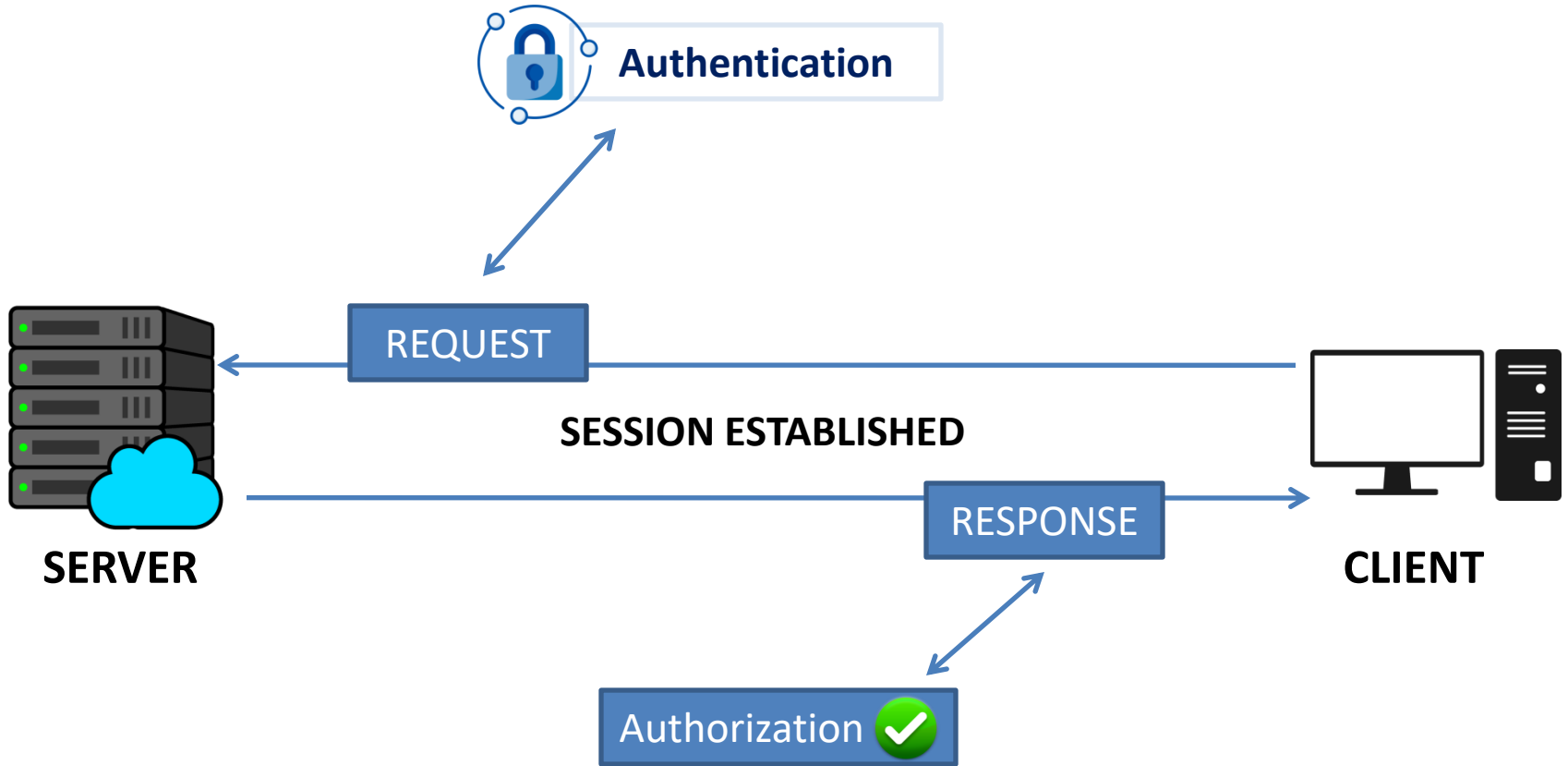
The tasks performed are:

- 1. Session Establishment, Maintenance, and Termination**
- 2. Synchronization**
- 3. Dialog Control**
- 4. Session Recovery**
- 5. Authentication, Authorization, and Accounting (AAA)**

Session Establishment, Maintenance, and Termination:

- **Establishment:** Initiates connections between applications on different devices, ensuring that communication sessions are properly started.
- **Maintenance:** Keeps the session alive and manages the data exchange, ensuring that the connection remains open and functional.
- **Termination:** Properly closes the session once communication is complete, freeing up resources.

SESSION LAYER:



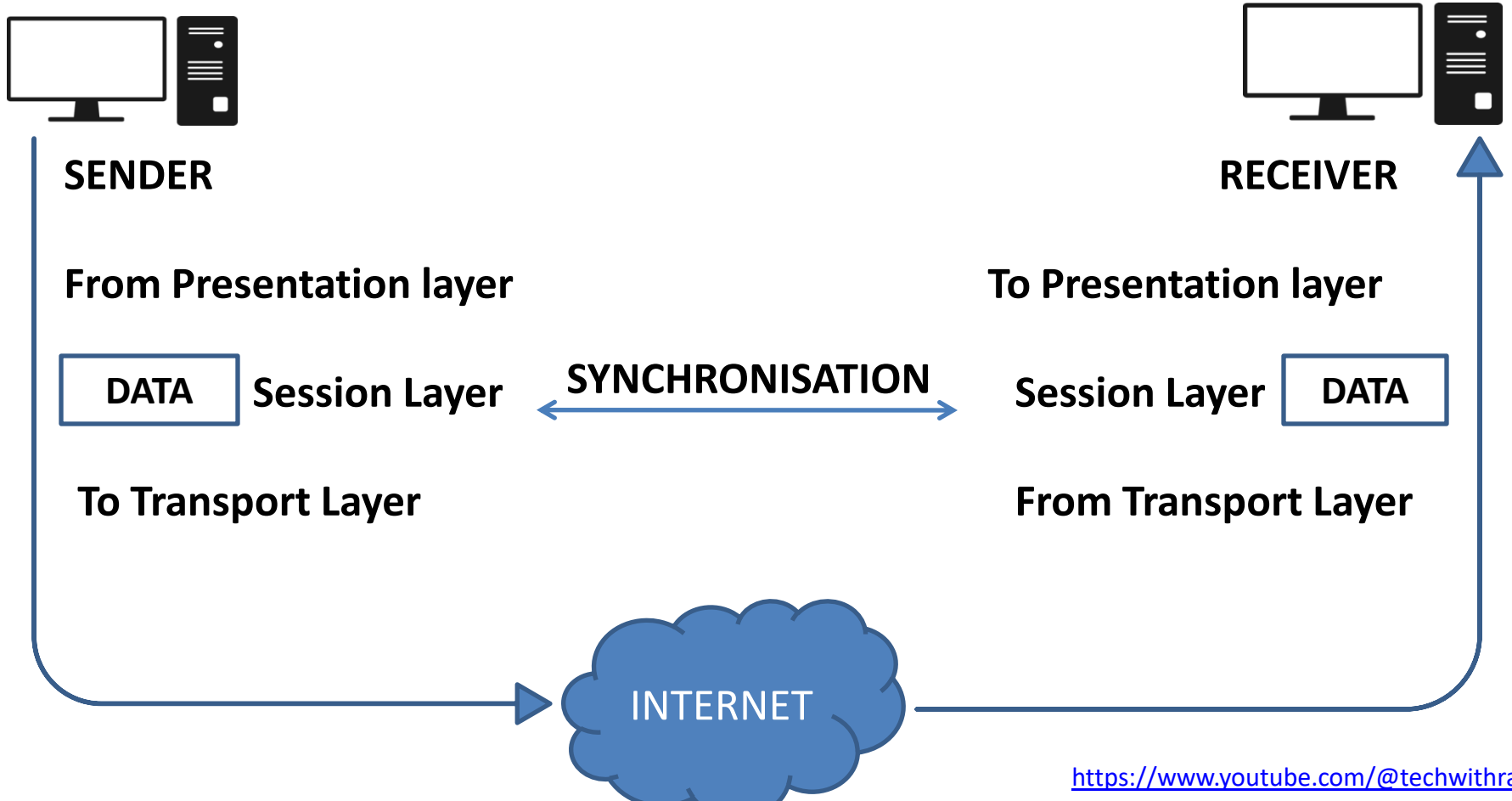
Synchronization:

- Manages synchronization points, or checkpoints, within a stream of data. If a session is interrupted, these checkpoints allow the session to be resumed from the last checkpoint rather than starting over.
- This ensures data integrity and continuity, particularly important for long communications or large data transfers.

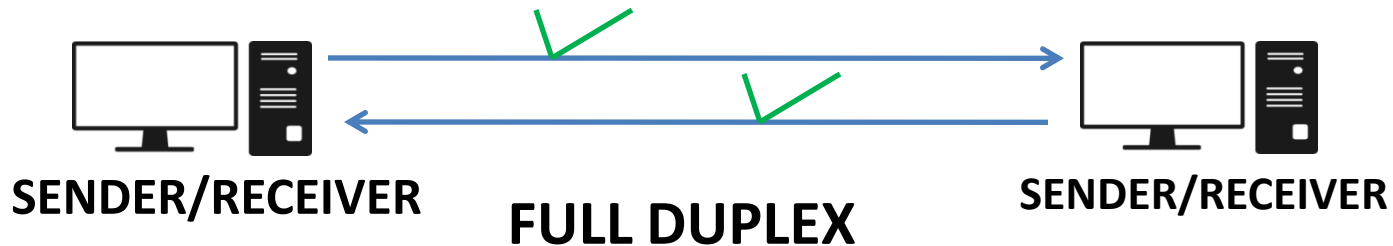
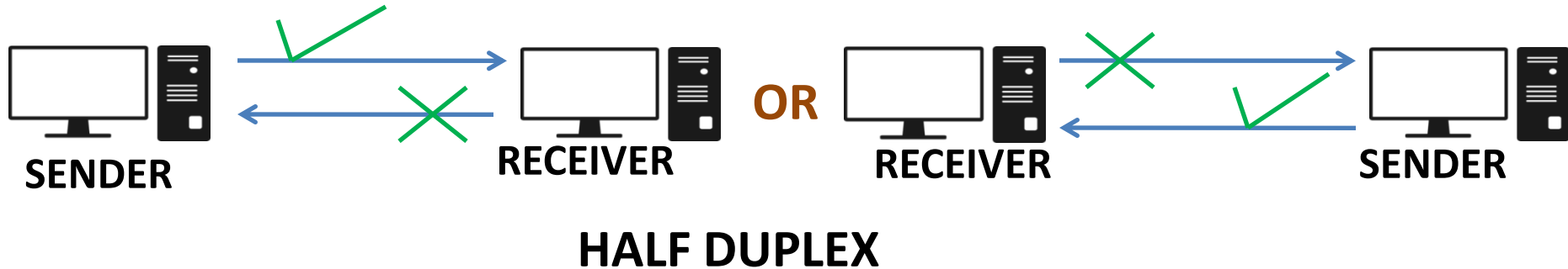
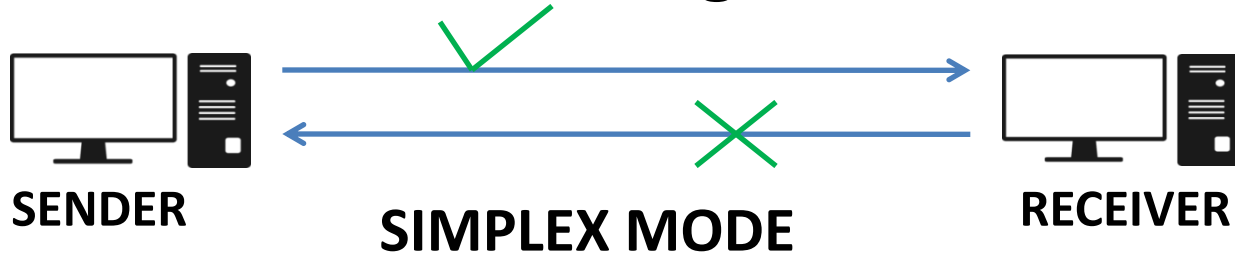
Dialog Control:

- Controls whether the communication is in half-duplex or full-duplex mode.
- **Half-Duplex:** Allows data transmission in one direction at a time.
- **Full-Duplex:** Allows simultaneous data transmission in both directions.

Session Layer:



Modes of Transfer or Dialog Control in Session Layer



Authentication, Authorization, and Accounting (AAA):

Authentication: Verifies the identity of users and devices attempting to establish a session. Ensures that only authorized entities can initiate communication.

Authorization: Determines whether an authenticated user has permission to access the requested resources or perform certain operations during the session.

Accounting: Tracks the usage of network resources by users and devices, often for auditing, billing, or resource allocation purposes.

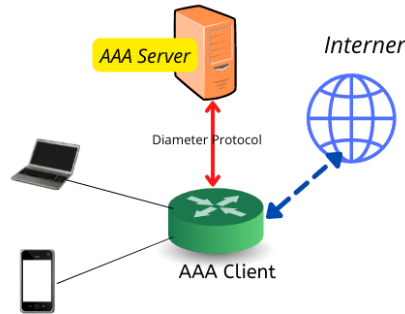
Authentication, Authorization, and Accounting (AAA):



Authentication



Authorization



Account